

PULS/e lab Eindhoven

Acoustic safety measurements

Research protocol attachment

Author: Vera van Hal
Version: 2
Date: 13 May 2022

Contents

1	Introduction.....	2
2	Limits	2
3	Methods	3
3.1	Mechanical	3
3.2	Thermal	5
4	Results	6
4.1	Mechanical	6
4.1.1	Dependence on transmit voltage	8
4.1.2	Dependence on center frequency	11
4.1.3	Dependence on transmit angle	14
4.1.4	Difference between transducers	17
4.1.5	Safety limits	17
4.2	Thermal measurements	20
4.2.1	“Free air” measurements	20
4.2.2	Test object measurements	21
4.2.3	Safety limits	24
5	Conclusions.....	24
6	Practical implementation of acoustic safety limits	25
	References	26
Appendix A	Acquisition script Verasonics.....	27

1 Introduction

In the Photoacoustics and Ultrasound Laboratory Eindhoven (PULS/e) at the Department of Biomedical Engineering of the Eindhoven University of Technology, several research studies are performed in which Vantage 256 systems (Verasonics, Kirkland, WA, USA) are used for ultrasound acquisition. In case this equipment is used for studies involving volunteers or patients, there needs to be proof that the equipment and developed acquisition schemes are safe and will not harm the test persons involved. In general, diagnostic ultrasound has an excellent safety record. However, exposure of tissues to intense levels of ultrasound, well above those levels found with diagnostic ultrasound devices, can have significant destructive effects [1]. Several Standards and Guidelines exist to give guidance on safety limits for ultrasound use (see Section 2).

All commercially available medical ultrasound systems are designed such that they remain well below the safety limits for mechanical and thermal properties. Moreover, Mechanical and Thermal Indices (MI and TI) are shown on the monitor enabling sonographers to keep an eye on the safe use of the system. Since research systems like the Verasonics enable researchers to program ultrasound beams, safety standards are not inherently met. Therefore, in order to design safe clinical studies with human subjects, information about the safety of the transmitted acoustic field is important.

At the PULS/e lab, mechanical and thermal measurements were performed to test the safety of the ultrafast acquisition schemes with the Verasonics C5-2v curved array transducer, that uses diverging wave transmits. This report describes the results of these measurements.

2 Limits

For different applications of ultrasound imaging, different mechanical and thermal safety limits exist. These safety limits are mentioned in [1] and [2]:

Neonatal, abdominal, intraoperative, pediatric, small organ, cephalic

	MI	I _{SPPA.3}	I _{SPTA.3}	T _{surface}	ΔT _{surface}
Limit	1.9	190 W/cm ²	94 mW/cm ²	43°C (test object) 50°C (free in air)	10°C (test object) 27°C (free in air)

Peripheral vessel

	MI	I _{SPPA.3}	I _{SPTA.3}	T _{surface}	ΔT _{surface}
Limit	1.9	190 W/cm ²	720 mW/cm ²	43°C (test object) 50°C (free in air)	10°C (test object) 27°C (free in air)

Cardiac

	MI	I _{SPPA.3}	I _{SPTA.3}	T _{surface}	ΔT _{surface}
Limit	1.9	190 W/cm ²	430 mW/cm ²	43°C (test object) 50°C (free in air)	10°C (test object) 27°C (free in air)

The absolute limit for the test object measurement (43°C) assumes a start temperature of 33°C. According to IEC 60601-2-37 (section 201.11.1.3.1.1.1) [2], one of the two test criteria can be selected. For these measurements, the test criterium for temperature rise (ΔT_{surface}) is used, provided that the initial temperature of the test object at the object-transducer interface is between 20°C and 33°C.

In addition to these hard limits, exposure time should be minimized.

3 Methods

3.1 Mechanical

A picture of the mechanical test set-up is shown in Figure 1. The test tank system was supplied from Precision Acoustics Ltd, Dorchester, UK. The test tank was filled with demi-water. A hydrophone (Onda HNP-0400) is used to measure acoustic pressure, and could be positioned at any desired measurement location using the UMS Control Software. The hydrophone, consisting of a needle and a pre-amplifier, is mounted in a holder that was in turn connected to the positioning system. The ultrasound transducer was also mounted in a custom made probe holder and was fully immersed in the water (see Fig. 2). An ultrasound probe cover (Civco) was used to prevent water ingress. Read-out of the pressure wave measured with the hydrophone was carried out using a Keysight InfiniiVision DSOX3024A oscilloscope and the UMS Control Software. The starting time of a measurement is coupled with the transmit event in the Verasonics via a trigger signal.

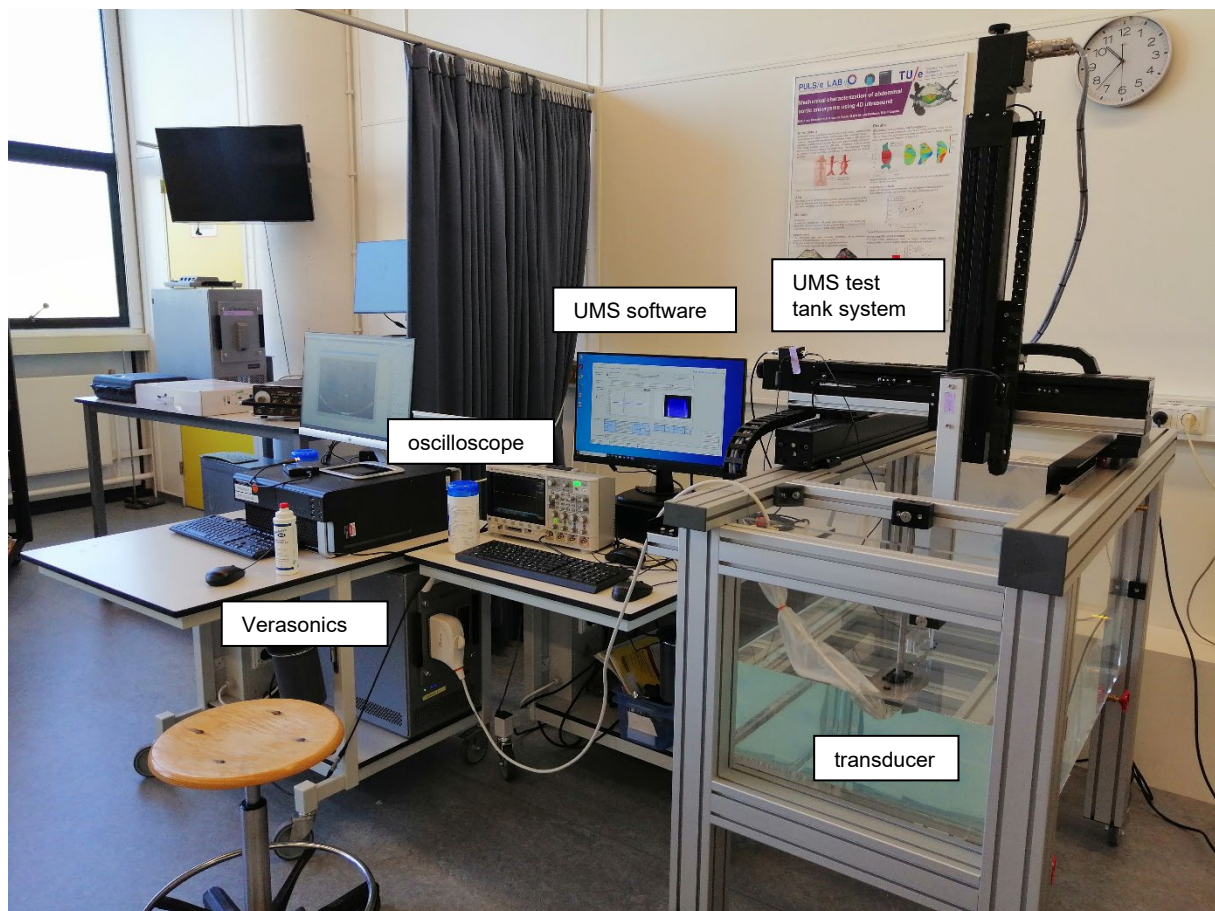


Figure 1 Hydrophone system overview

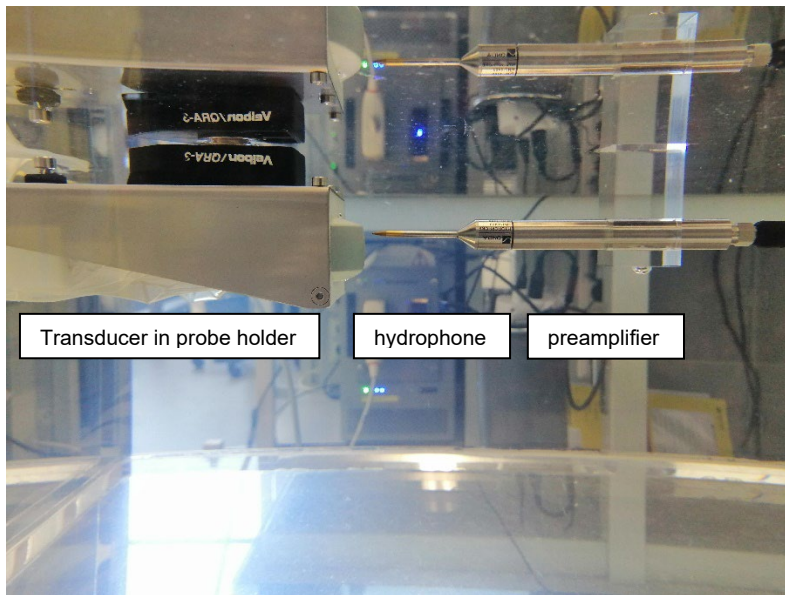


Figure 2 Hydrophone and ultrasound transducer submerged in water.

The hydrophone measures the acoustic output by means of a voltage. With use of a calibration table this voltage can be converted into a pressure. Data processing could be performed offline, using the Post Processing Software from Precision Acoustics. The main parameters computed are:

- Acoustic working frequency (f_{awf})
- Pulse-intensity integral (pii)
- Spatial-peak pulse-average intensity (I_{SPPA})
- Spatial-peak temporal-average intensity (I_{SPTA})
- Mechanical Index (MI)

The measurements are carried out inside the water tank. Since the attenuation of the ultrasound signal in human tissue differs from that in water, the results need to be compensated for this difference. An attenuation coefficient of $0,3 \text{ dB cm}^{-1}\text{MHz}^{-1}$ is used for this purpose. Derated parameters are denoted with a subscript “.3”. International standard IEC 62359 [3] describes the official method to derive the above parameters. The manual of the Post Processing Software [4] describes how this is performed in the software used.

Default measurement settings are:

- Diverging wave
- Averaging over 32 measurements per measurement location (Oscilloscope/UMS software)
- Time scale: approximately $2 \mu\text{s}/\text{Div}$ for XY scans (Oscilloscope)
- Output voltage: 30 V (Verasonics)
- Center frequency: 3.6765 MHz (C5-2v)
- Number of pulses: 1 per transmit burst (Verasonics)
- Pulse repetition frequency: 1 kHz (TimeToNextAcq 1000 μs , Verasonics)
- Number of active transducer elements: 128 (C5-2v)
- Wave angle: 0°

Several of these parameters are varied during the series of measurements, which are the transmit voltage, transmit angle and center frequency. This will be described in Section 4.1. All measurements were carried out with an appropriate transmit apodization to obtain a homogeneous beam. The Verasonics script used for these measurements can be found in Appendix A.

At least one measurement is performed that compromises the entire transmitted acoustic field, starting close to the transducer surface to the location of the elevational focus. For studying the impact of the varying settings, smaller measurement ranges are chosen in order to limit measurement time.

When carrying out these experiments some points of attention need to be kept in mind. The hydrophone tip should be aligned with the beam axis of the transducer and the dimensions of the water tank. Any air bubbles between the ultrasound gel and the probe cover should be removed. Water temperature needs to be measured before every measurement, and if necessary be adapted in the UMS Control Software on the computer. Moreover, choosing the correct time range and scale at the oscilloscope is crucial for reliable measurements.

3.2 Thermal

In order to test the surface temperature of the transducer, two set-ups were built:



Figure 3 Test set-up to measure surface temperature of the ultrasound transducer in free air. A thermocouple is fixated to the center of the transducer surface. Additionally, the temperature is measured with a Fluke IR camera.



Figure 4 Test set-up to measure surface temperature of the ultrasound transducer on a test object. A CIRS phantom (Model 539) is used as the test object. A thermocouple was attached to the probe surface. Additionally, the temperature was measured with a Fluke IR camera.

One set-up is used to measure the temperature in “free air” (Fig. 3), and the other set-up is used to measure the temperature in a tissue-mimicking test object (Fig. 4). The temperature was measured using two types of temperature probes: a thermocouple (voltcraft K202 datalogger) and an IR camera (Fluke 62 max). The transducer transmits the same acoustic pulses as used in the mechanical tests with the hydrophone.

4 Results

4.1 Mechanical

Measurements were performed by Vera van Hal between 25 October 2021 and 9 December 2021. A typical signal as measured by the hydrophone is shown in Figure 5.

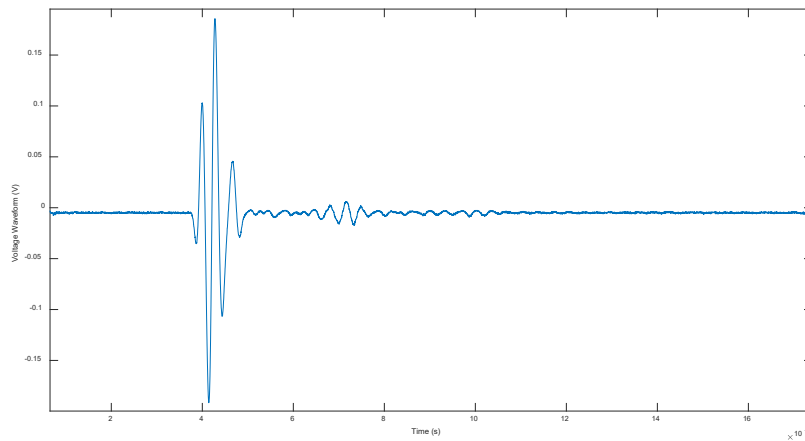


Figure 5 Waveform measured by the hydrophone with default transmit settings.

With the use of the post processing software, the relevant parameters can be computed for every measurement location within the measurement grid.

An impression of the acoustic output field of the C5-2v transducer is given by plotting the pii and pii_3 in the XZ-plane, and the YZ-plane in Fig. 6 and Fig. 7, respectively.

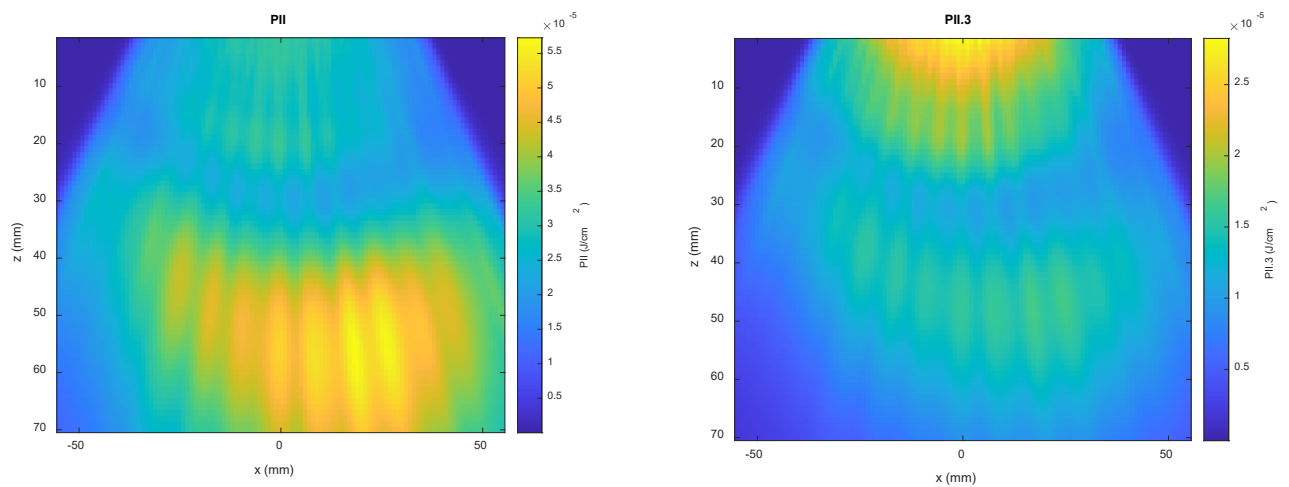


Figure 6 Pulse Intensity Integral (pii) and its attenuated counterpart (pii_3) of the C5-2v transducer in the xz-plane at the elevational center of the transducer. Including the higher attenuation by tissue than by water, the location of maximum intensity shifts from the elevational focus at $z=60\text{mm}$ to nearby the transducer.

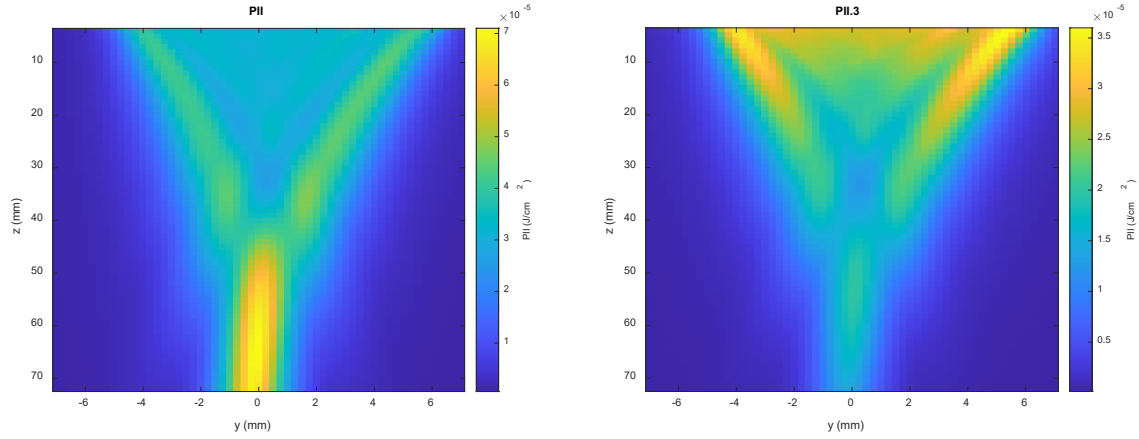


Figure 7 Pulse Intensity Integral (pii) and its attenuated counterpart (pii.3) of the C5-2v transducer in the yz-plane at the lateral center of the transducer. Including the higher attenuation by tissue than by water, the location of maximum intensity shifts from the elevational focus at z=60mm to nearby the transducer.

It is shown that the position of maximum intensity is shifted from the elevational focus to closer to the transducer when tissue attenuation is incorporated. Parameters MI and $I_{SPPA,3}$ show similar pictures. The pii close to the transducer surface in the XY-plane and its attenuated counterpart is plotted in Fig. 8.

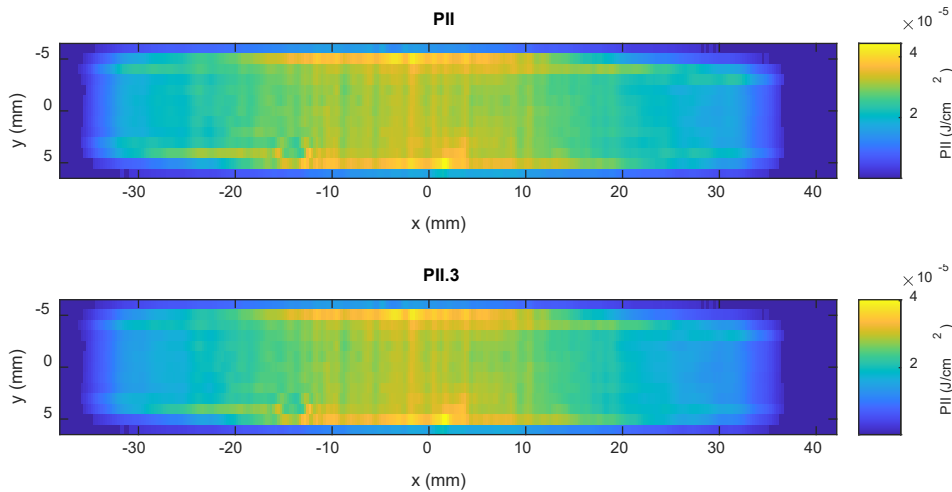


Figure 8 Pulse intensity integral (pii) and its attenuated counterpart (pii.3) of the C5-2v transducer in the xy-plane at a distance of z=2mm from the transducer surface at x=0mm.

It can be seen that the maximum intensity is found at the elevational edges of the transducer. The intensity decreases towards the lateral sides of the transducer, because of the increasing incidence angle with the hydrophone tip. A directivity correction based on the sensitivity curve of the hydrophone is used to obtain the homogeneous transmitted beam.

According to the Onda hydrophone handbook [5], the directivity follows the following formula:

$$D(\theta, f) = \left(\frac{1 + \cos \theta}{2} \right) * \left(\frac{2 J_1(ka \sin \theta)}{ka \sin \theta} \right)$$

Where:

D = directivity function (voltage out of the hydrophone relative to voltage when an incoming wave is normal to the tip)

Θ = the angle relative to the normal

f = frequency

J_1 = Bessell function of the first kind

k = wave number ($2\pi / \text{wavelength}$) = $2\pi f/c$, where c is the wave speed (1485 m/s in water of 21°C [6])

a = radius of the hydrophone (0.2 mm for a HNP-0400 hydrophone).

The directivity of the HNP-0400 hydrophone as function of the incidence angle for a center frequency of 3.6765 MHz is plotted below:

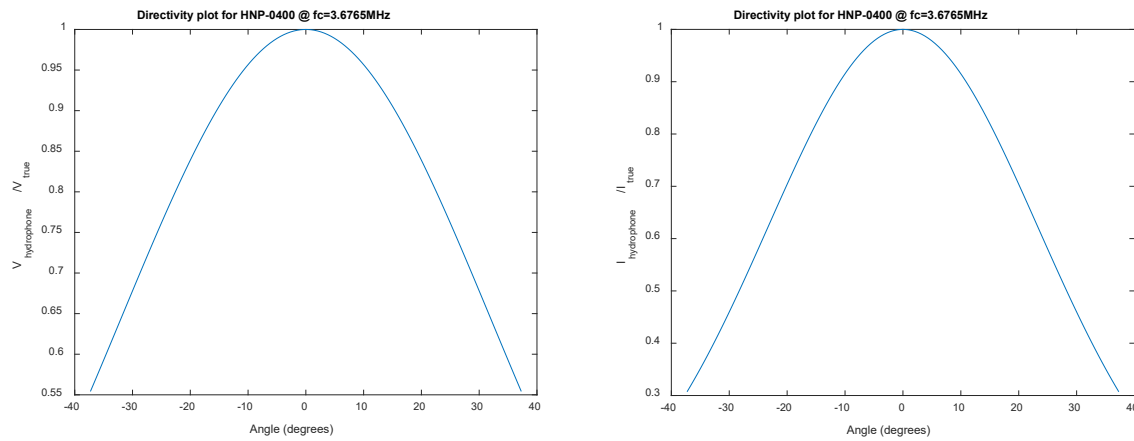


Figure 9 Directivity of the HNP-0400 hydrophone as a function of the incidence angle, for a center frequency of 3.6765MHz. (Left) Relation between the incidence angle and the measured output voltage. (Right) Relation of the incidence angle and the measured output intensity.

It can be seen that for the outer transducer elements, the measured output voltage, and thereby acoustic pressure is 45% less than what is truly transmitted. The measured output intensity scales quadratically with the measured output voltage.

The directivity plot is projected onto the measured plane at a certain distance from the transducer surface to obtain the correction matrix (1/directivity), which could be multiplied with the measured pressure/intensity. The corrected $p_{ii,3}$ in the XY plane is shown in Fig. 10. From this plot, global maximum $p_{ii,3}$ was reported for every measurement. Similarly, global maximum MI and $I_{SPPA,3}$ values were also obtained from their plots after directivity correction.

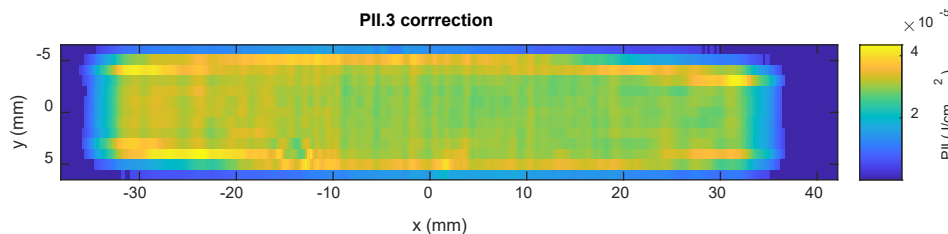


Figure 10 Attenuated pulse intensity integral ($p_{ii,3}$) after directivity correction.

4.1.1 Dependence on transmit voltage

Several parameters are varied throughout the series of experiments. The influence of output voltage and on the most relevant parameters from a safety perspective – MI, $I_{SPPA,3}$ and $p_{ii,3}$ – is shown in Fig. 11 until Fig. 13.

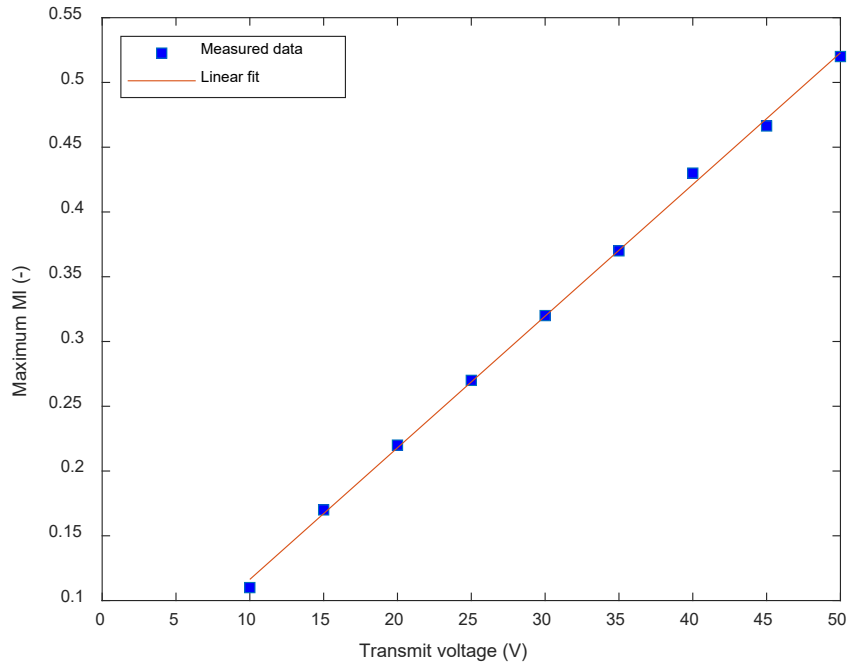


Figure 11 Maximum mechanical index (MI) as a function of transmit voltage. Default settings for center frequency ($f_c=3.6765$ MHz) and steering angle (0°) were used. Safety limit is 1.9.

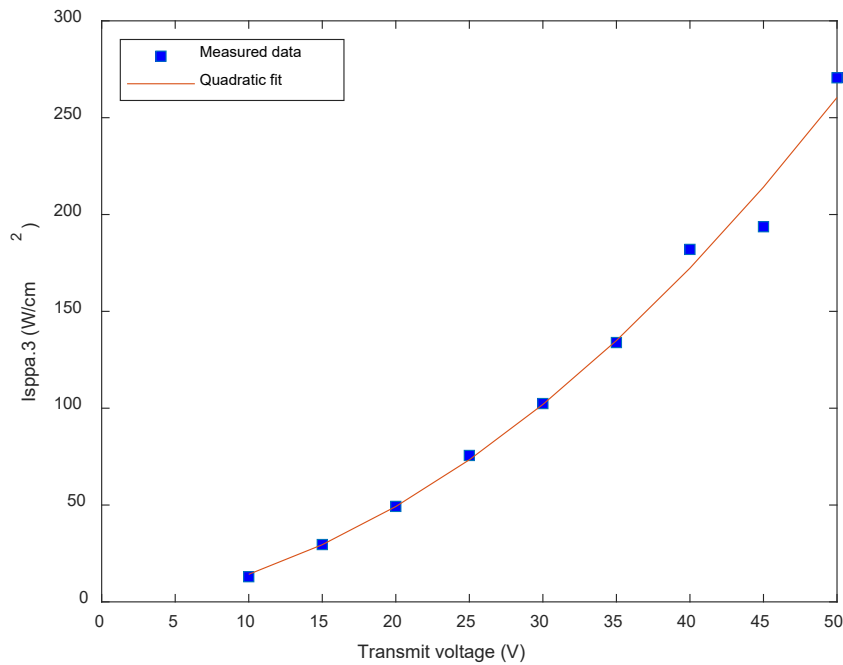


Figure 12 Attenuated Spatial Peak Pulse Average Intensity ($I_{SPPA.3}$) as a function of transmit voltage. Default settings for center frequency ($f_c=3.6765$ MHz) and steering angle (0°) were used. Safety limit is 190 W/cm^2 .

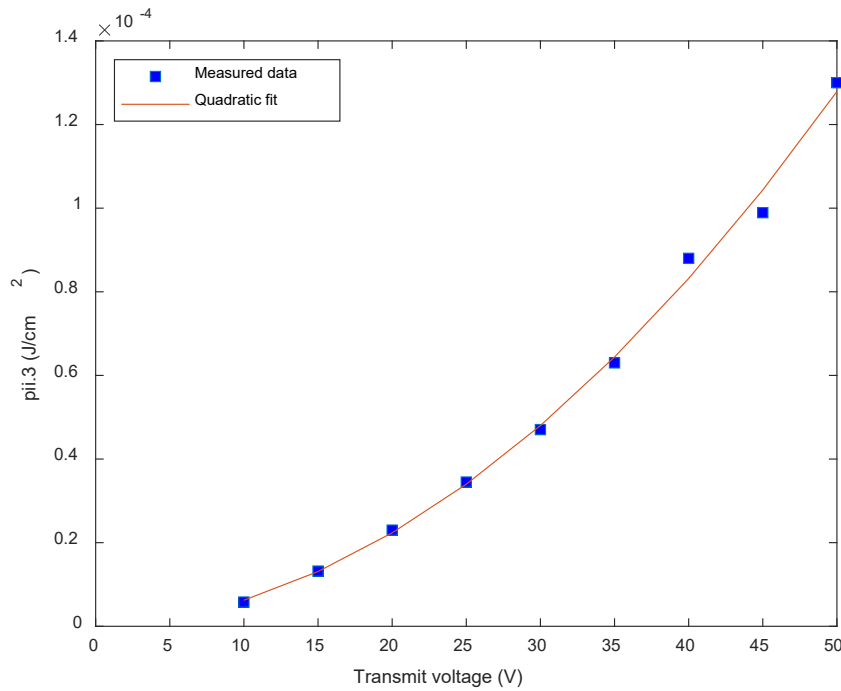


Figure 13 Attenuated pulse intensity integral ($pii.3$) as a function of transmit voltage. Default settings for center frequency ($f_c=3.6765$ MHz) and steering angle (0°) were used.

The mechanical index has a linear relation with the acoustic pressure, and thereby also the transmit voltage. This corresponds to the findings in Fig 11. The $I_{SPPA.3}$ and $pii.3$ scale quadratically with acoustic pressure. Therefore, the measured data points in Fig. 12 and Fig. 13 match with a quadratic fit.

According to the calibration document, the uncertainty of the hydrophone measurement is $\pm 7\%$. Since this is below 15%, no additional uncertainty is taken into account, as stated by the FDA [1]. A summary of the safety measurements is presented in the table below. It is shown that for the default settings, all relevant parameters are below the safety limits if the transmit voltage does not exceed 40V.

Table 1 Maximum MI, $I_{SPPA.3}$ and $pii.3$ measurements for the C5-2v transducer. Results are shown for default settings of the center frequency ($f_c=3.6765$ MHz) and steering angle (0°).

Transmit voltage (V)	MI	$I_{SPPA.3}$ (W/cm ²)	$pii.3$ (J/cm ²)
10V	0.11	13.0	$5.8 * 10^{-6}$
15V	0.17	29.6	$1.3 * 10^{-5}$
20V	0.22	49.3	$2.3 * 10^{-5}$
25V	0.27	75.6	$3.5 * 10^{-5}$
30V	0.32	102.4	$4.7 * 10^{-5}$
35V	0.37	133.9	$6.3 * 10^{-5}$
40V	0.43	182.0	$8.8 * 10^{-5}$
45V	0.47	193.7 (above limit)	$9.9 * 10^{-5}$
50V	0.52	270.6 (above limit)	$1.3 * 10^{-4}$

4.1.2 Dependence on center frequency

The influence of center frequency on the most relevant parameters from a safety perspective – MI, $I_{SPPA,3}$ and $p_{ii,3}$ – is shown in Fig 14 until Fig 16. The tested center frequencies are chosen such that the sampling rate is consistent with the internal clock of the Verasonics.

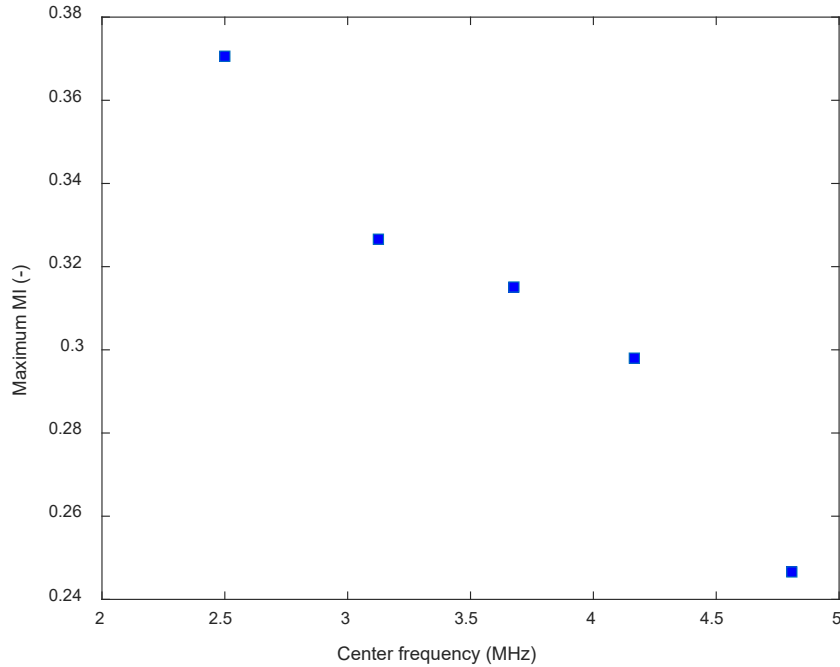


Figure 14 Mechanical index (MI) as a function of center frequency. Default settings for transmit voltage (30 V) and steering angle (0°) were used. Safety limit is 1.9.

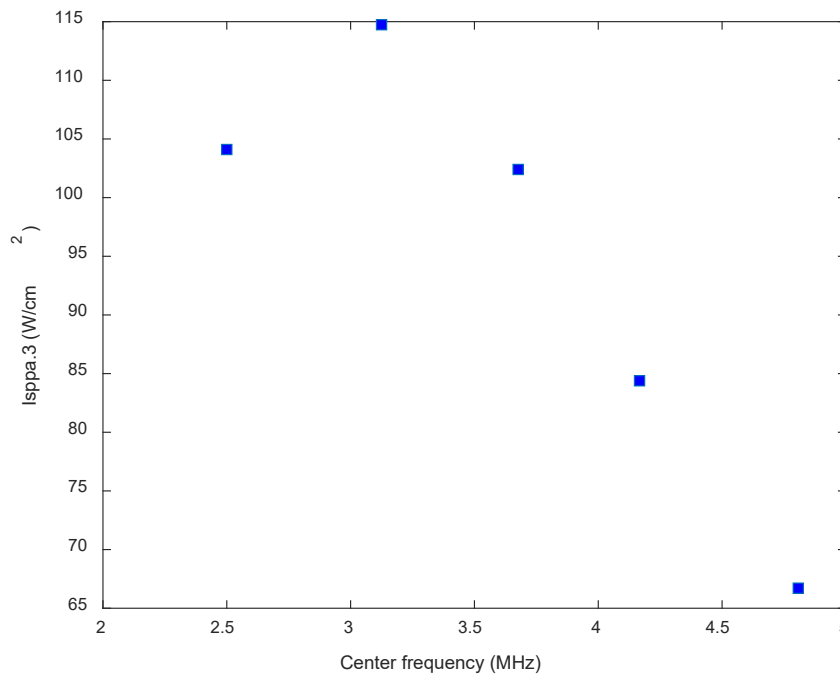


Figure 15 Attenuated Spatial Peak Pulse Average Intensity ($I_{SPPA,3}$) as a function of center frequency. Default settings for transmit voltage (30 V) and steering angle (0°) were used. Safety limit is 190 W/cm².

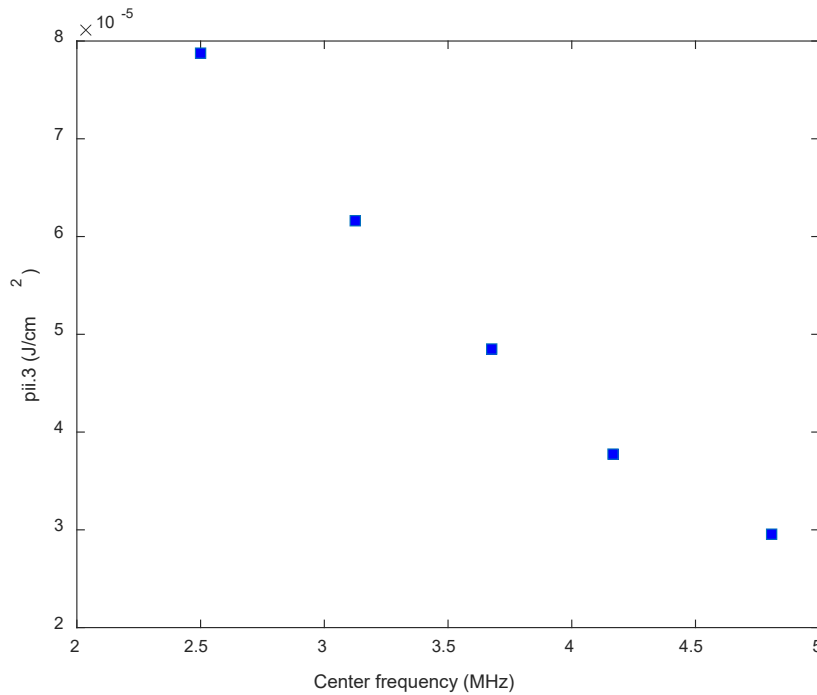


Figure 16 Attenuated pulse intensity integral (pii.3) as a function of center frequency. Default settings for transmit voltage (30 V) and steering angle (0°) were used.

The $I_{SPPA,3}$ shows a maximum for a center frequency of 3.125 MHz. The transducer elements prefer to vibrate at this frequency. The MI and pii.3 show a decrease of acoustic output with increasing center frequency, because higher frequencies are attenuated more than lower frequencies.

We also studied the relationship between the center frequency and the acoustic working frequency. It is shown that the optimal resonance frequency of the C5-2v transducer is around 3 MHz, which also corresponds with the peak in $I_{SPPA,3}$. At a center frequency of 2.5 MHz, the acoustic working frequency was found to be equal to the center frequency. However, above 3 MHz, the acoustic working frequency is found to be lower than the center frequency.

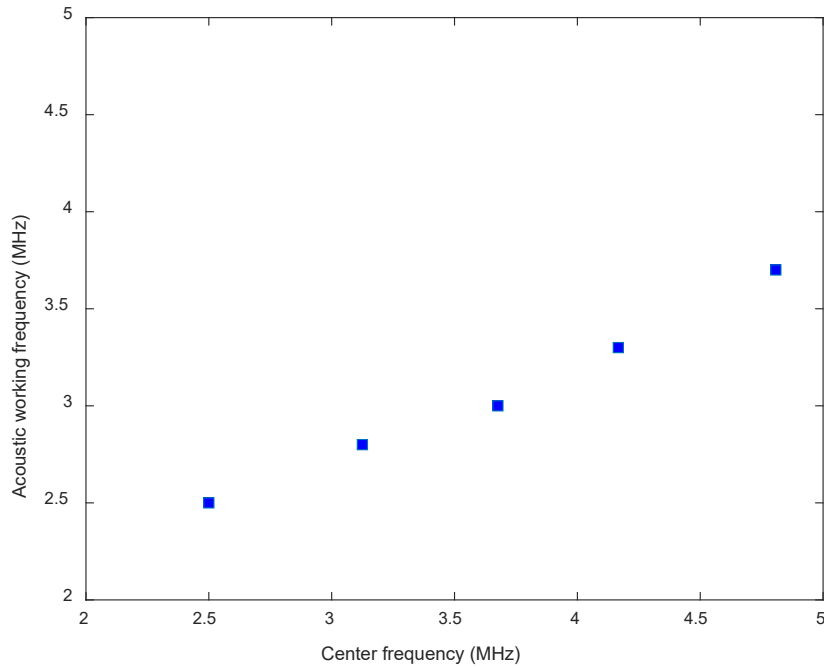


Figure 17 Relationship between center frequency and acoustic working frequency. Default settings for transmit voltage (30 V) and steering angle (0°) were used.

A summary of the safety measurements is presented in the tables below. It is shown that for the default settings, all relevant parameters are below the safety limits for all tested center frequencies.

Table 2 Maximum MI, $I_{SPPA,3}$ and $p_{ii,3}$ measurements for the C5-2v transducer. Results are shown for default settings of the transmit voltage (30 V) and steering angle (0°).

Center frequency (MHz)	MI	$I_{SPPA,3}$ (W/cm ²)	$p_{ii,3}$ (J/cm ²)
2.5 MHz	0.37	104.1	$7.9 \cdot 10^{-5}$
3.125 MHz	0.33	114.7	$6.2 \cdot 10^{-5}$
3.6765 MHz	0.32	102.4	$4.7 \cdot 10^{-5}$
4.1667 MHz	0.30	84.4	$3.8 \cdot 10^{-5}$
4.8077 MHz	0.25	66.7	$3.0 \cdot 10^{-5}$

4.1.3 Dependence on transmit angle

The pii and $pii_{.3}$ in the XZ-plane under a steering angle of 10° is plotted in Fig. 18.

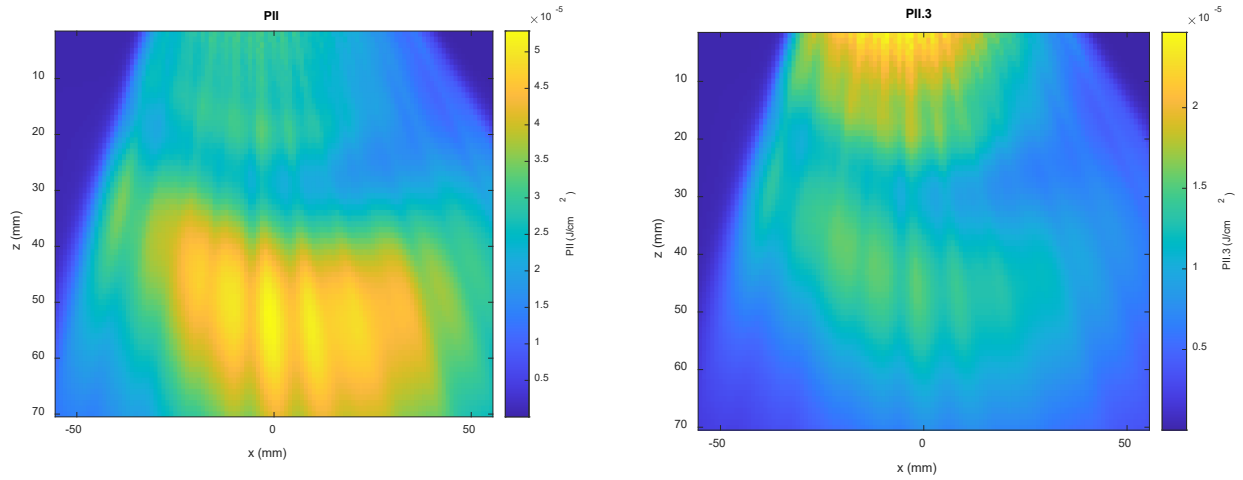


Figure 18 Pulse Intensity Integral (pii) and its attenuated counterpart ($pii_{.3}$) of the C5-2v transducer in the XZ-plane at the elevational center of the transducer. A steering angle of 10° is applied. Note that no directivity correction is applied in the visualization of this transmitted acoustic field.

The transmit angle is defined as the angle between the virtual source of the 0° angle, and the current virtual source, at a distance of 49.57mm (transducer radius) from the center element. To correct for the directivity curve of the hydrophone under a certain steering angle, the calculation of the incidence angle with the hydrophone tip, in the scanning plane close to the transducer surface, is now based on this shifted location of the virtual source. The result for a 10° steering angle is shown in Fig. 19 below.

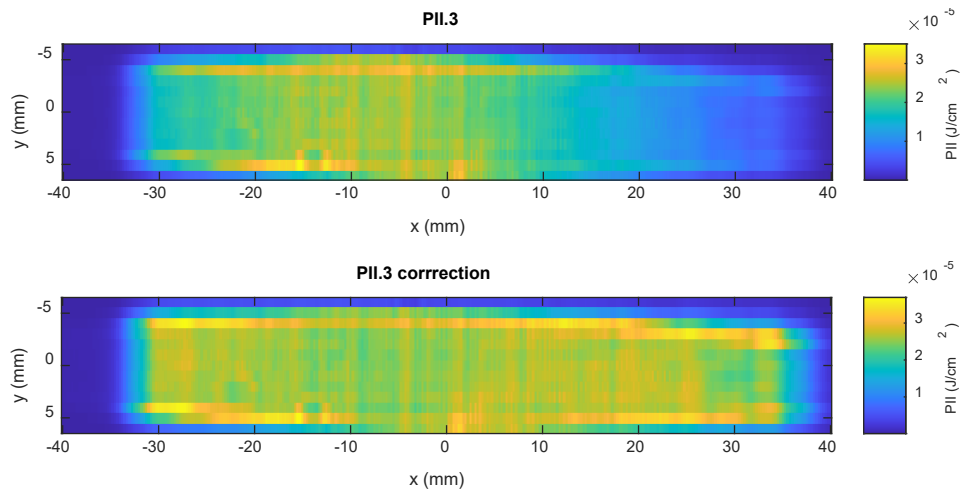


Figure 19 Attenuated pulse intensity integral ($pii_{.3}$) in the xy-plane close to the transducer surface. A 10° steering angle is applied. Default settings for transmit voltage (30 V) and center frequency ($f_c=3.6765$ MHz) were used. Top: Measured $pii_{.3}$ without directivity correction. Bottom: Measured $pii_{.3}$ after directivity correction.

The influence of the transmit angle on the most relevant parameters from a safety perspective – MI, $I_{SPPA.3}$ and $pii_{.3}$ – is shown in Fig. 20 until Fig. 22.

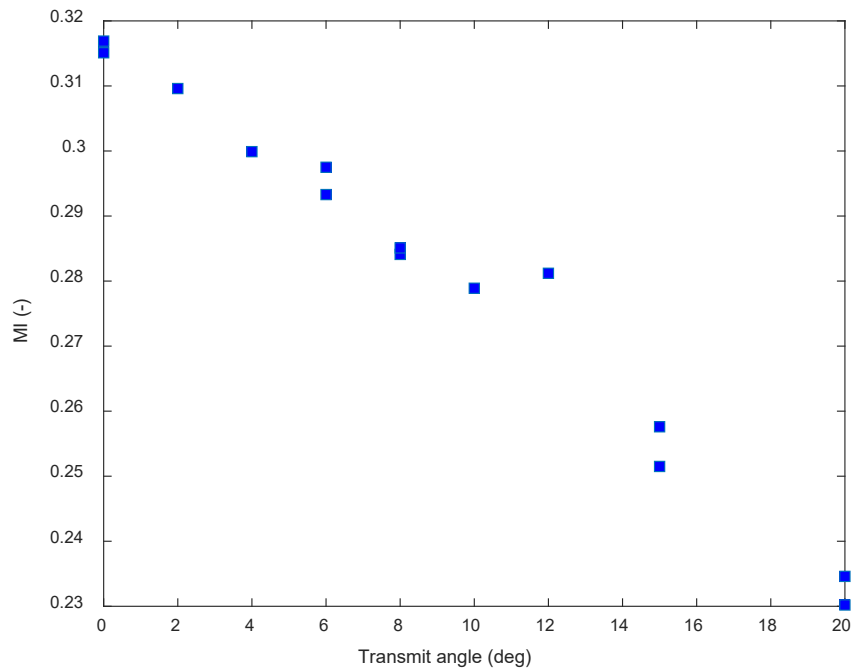


Figure 20 Mechanical index (MI) as a function of transmit angle ($^{\circ}$). Default settings for transmit voltage (30 V) and center frequency ($f_c = 3.6765$ MHz) were used. Safety limit is 1.9.

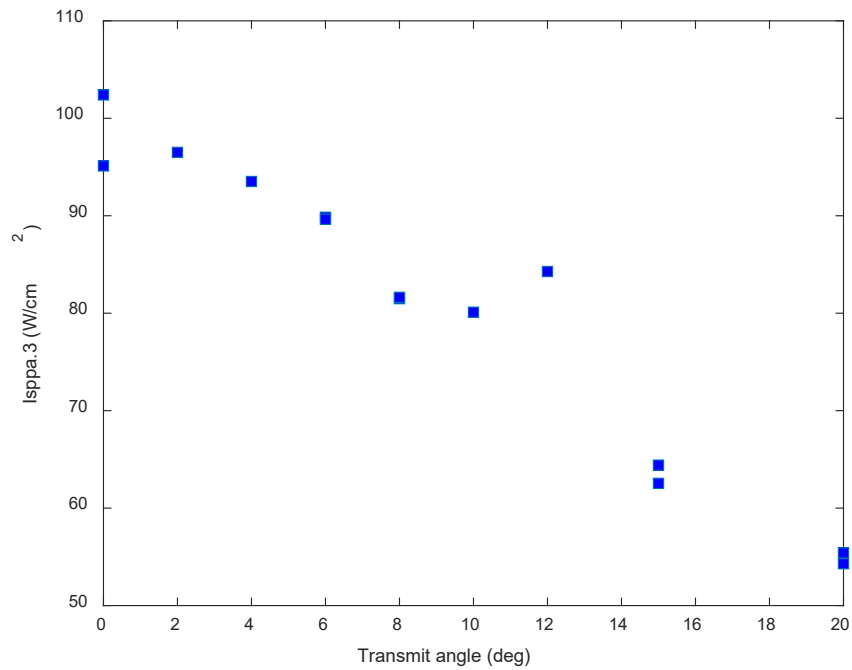


Figure 21 Attenuated Spatial Peak Pulse Average Intensity ($I_{SPPA.3}$) as a function of transmit angle ($^{\circ}$). Default settings for transmit voltage (30 V) and center frequency ($f_c = 3.6765$ MHz) were used. Safety limit is 190 W/cm 2 .

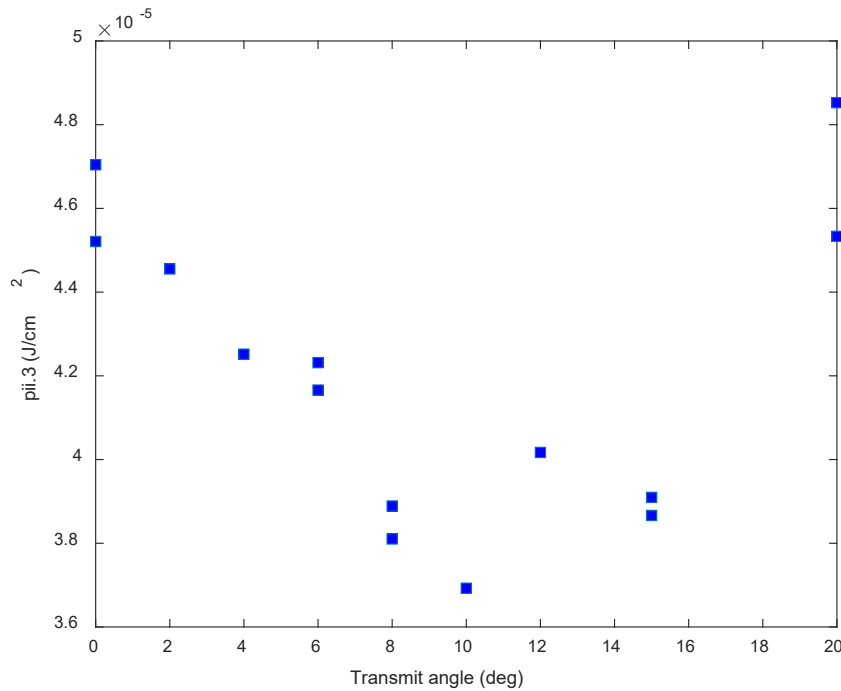


Figure 22 Attenuated pulse intensity integral (pii.3) as a function of transmit angle (°). Default settings for transmit voltage (30 V) and center frequency ($f_c = 3.6765$ MHz) were used.

It is shown that the MI and $I_{SPPA,3}$ decrease with increasing transmit angle. I_{SPPA} , pii and the computed pulse duration are based on the range between 10% and 90% of the cumulative intensity. With a certain transmit angle, it can be seen that an additional tail is present behind the main peak in the waveform (see Fig. 23). This tail is caused by the different time delays between the individual transducer elements. The hydrophone is hit by the waves coming from multiple transducer elements. Because considerable energy is present in this tail, the computed pulse duration increases, leading to a decrease in I_{SPPA} .

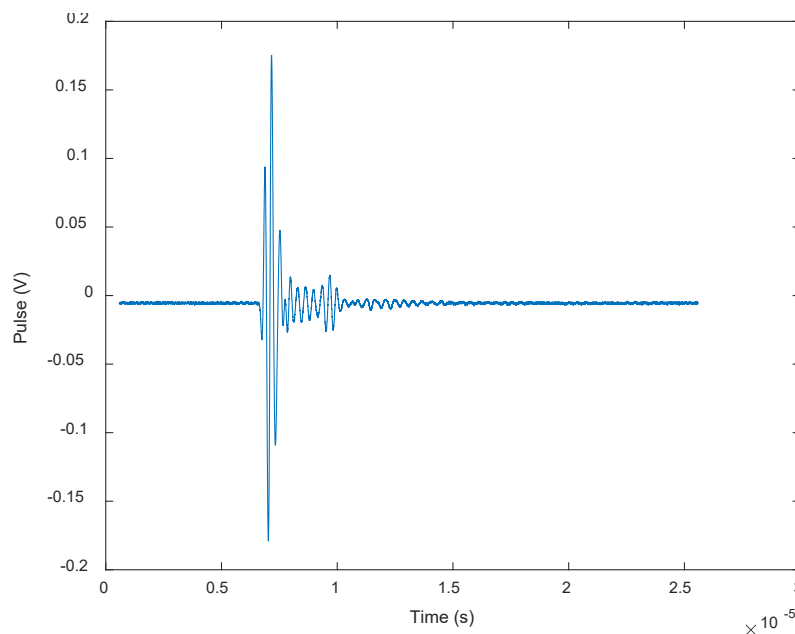


Figure 23 Waveform measured with the hydrophone with a transmit angle of 10°.

The maximum $pii_{.3}$ also decreased with transmit angles $< 10^\circ$, which can be explained by the directivity of the transducer elements. With transmit angles $> 10^\circ$, maximum $pii_{.3}$ increased, while the mean $pii_{.3}$ over the whole scanning region was still found to decrease. The increase in $pii_{.3}$ at some positions can be explained by a small increase of energy in the tail, as individual waves from more elements can reach these locations with large transmit angles.

A summary of the safety measurements is presented in the table below. If multiple measurements were performed for one transmit angle, the largest value is reported. It is shown that for the default settings, all relevant parameters are below the safety limits for all transmit angles.

Table 3 Maximum MI, $I_{SPPA.3}$ and $pii_{.3}$ measurements for the C5-2v transducer. Results are shown for default settings of the center frequency ($f_c=3.6765$ MHz) and transmit voltage (30V).

Transmit angle (deg)	MI	$I_{SPPA.3}$ (W/cm ²)	$pii_{.3}$ (J/cm ²)
0°	0.32	102.4	$4.8 * 10^{-5}$
2°	0.31	96.5	$4.5 * 10^{-5}$
4°	0.30	93.5	$4.3 * 10^{-5}$
6°	0.30	89.8	$4.2 * 10^{-5}$
8°	0.29	81.6	$3.8 * 10^{-5}$
10°	0.28	80.1	$3.7 * 10^{-5}$
12°	0.28	84.3	$4.0 * 10^{-5}$
15°	0.26	64.4	$3.9 * 10^{-5}$
20°	0.23	55.4	$4.9 * 10^{-5}$

4.1.4 Difference between transducers

The measurements were performed with the C5-2v transducer with serial number 43015795. To test the variation between the transducer, and between measurements, the default settings were tested multiple times. The differences between measurements are tabulated below:

Table 4 Maximum MI, $I_{SPPA.3}$ and $pii_{.3}$ measurements for the C5-2v transducer. Results are shown for default settings of the transmit voltage (30V), center frequency ($f_c=3.6765$ MHz) and steering angle (0°).

Transducer serial number	MI	$I_{SPPA.3}$ (W/cm ²)	$pii_{.3}$ (J/cm ²)
43015795	0.32	102.4	$4.8 * 10^{-5}$
43015795	0.32	95.1	$4.5 * 10^{-5}$
79048288	0.31	91.1	$4.3 * 10^{-5}$

All these measurements fall within the $\pm 7\%$ uncertainty range reported in the calibration document of the hydrophone. It can be concluded that there are no significant differences between the transducers and the measurements.

4.1.5 Safety limits

Based on the safety limit for the $I_{SPTA.3}$ of 94 mW/cm², the maximum allowed pulse repetition frequency (PRF) can be calculated from the relationship between $pii_{.3}$ and $I_{SPTA.3}$:

$$I_{SPTA.3} = pii_{.3} * PRF.$$

Based on Fig. 22, the maximum allowed PRF for each transmit angle is plotted in Fig. 24 and tabulated in Table 5.

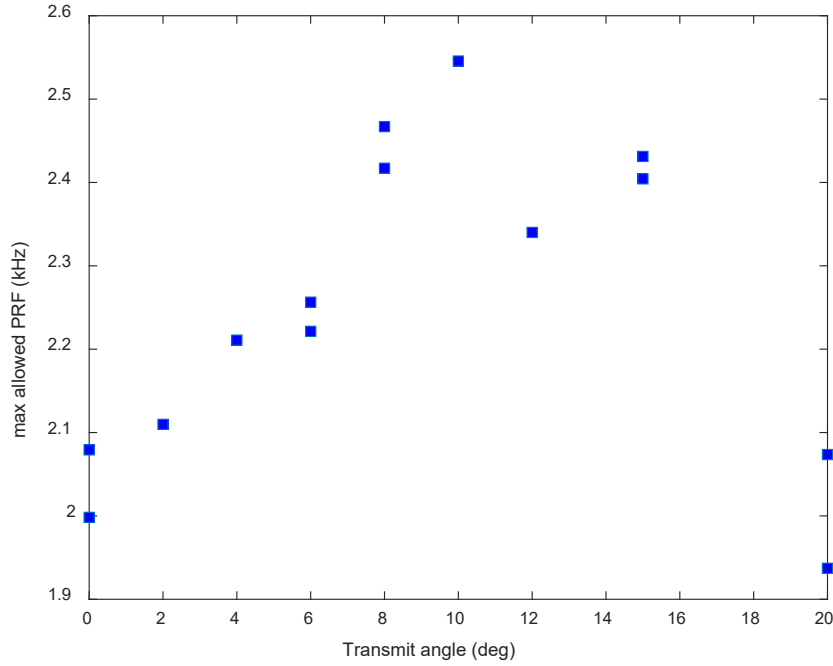


Figure 24 Maximum allowed PRF, considering the measured values for $pii.3$ in Fig 22, and the $I_{SPTA.3}$ limit of 94 mW/cm² for neonatal, abdominal, intraoperative, pediatric, small organ and cephalic procedure applications, as a function of transmit angle.

Table 5 Maximum PRF for different transmit angles. PRF values are based on the measured maximum values for $pii.3$ in Table 3 and the FDA limit for $I_{SPTA.3}$.

Transmit angle (deg)	PRF max (kHz)
0°	2.0
2°	2.1
4°	2.2
6°	2.3
8°	2.4
10°	2.5
12°	2.3
15°	2.4
20°	1.9

In theory, the maximum allowed PRF for a scanning sequence can be calculated from the effective $pii.3$, which is defined by:

$$effective\ pii.3 = \frac{1}{N_A} \sum_{n=1}^{N_A} (pii.3)_n$$

where N_A is the number of angles in the sequence. The measurements were performed with a PRF of 1 kHz, and a transmit voltage of 30V. Since $pii.3$ is also a function of output voltage (see Fig. 13), the maximum acceptable PRF also depends on V. The maximum PRF for a given V, at the 0° angle is shown in Fig. 25.

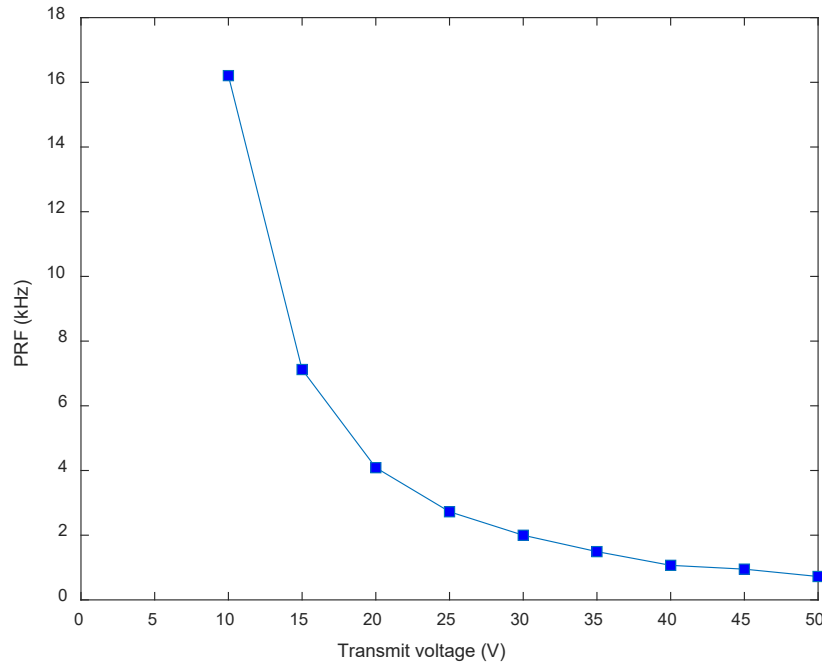


Figure 25 Maximum allowed PRF, considering the measured values for $pii.3$ in Fig 13, and the $I_{SPTA.3}$ limit of 94 mW/cm² for neonatal, abdominal, intraoperative, pediatric, small organ and cephalic procedure applications, as a function of transmit voltage.

The maximum allowed PRF for varying transmit voltage is tabulated below:

Table 6 Maximum PRF for different transmit voltages. PRF values are based on the measured maximum $pii.3$ values in Table 1 and the FDA limit for $I_{SPTA.3}$.

Transmit voltage (V)	PRF max (kHz)
10V	16
15V	7.1
20V	4.1
25V	2.7
30V	2.0
35V	1.5
40V	1.1
45V	0.95
50V	0.70

To check this approach, a 20V acquisition was performed with a PRF of 4kHz. The resulting $I_{SPTA.3}$ was equal to 84.5 mW/cm², which is indeed below the safety limit.

Since the maximum allowed PRF is increased with a steering angle $> 0^\circ$, it is safe to say that the maximum allowed PRF in Table 6 is a conservative limit for each scanning sequence using that voltage. For intermediate voltages, the $pii.3$ and thereby maximum allowed PRF can be calculated according to the polynomial fit in Fig. 13:

$$PII.3 = 4,7 \cdot 10^{-8} V^2 + 2,3 \cdot 10^{-7} V - 8,6 \cdot 10^{-7}$$

In practice, the maximum achievable PRF is about 7 kHz, because of the large imaging depth.

Similarly, the maximum allowed PRF for varying center frequency is tabulated below:

Table 7 Maximum allowed PRF for different center frequencies. PRF values are based on the measured maximum $p_{ii,3}$ in Table 2, and the FDA limit for $I_{SPTA,3}$.

Center frequency (MHz)	PRF max (kHz)
2.5 MHz	1.2
3.125 MHz	1.5
3.6765 MHz	2.0
4.1667 MHz	2.5
4.8077 MHz	3.2

4.2 Thermal measurements

Measurements were performed by Vera van Hal and Hans-Martin Schwab on Friday 3 December 2021 and Monday 6 December 2021.

4.2.1 “Free air” measurements

In Fig. 26, the surface temperature was measured over time using the default transmit settings in free air. Infrared (IR) and thermocouple (TC) measurements were compared.

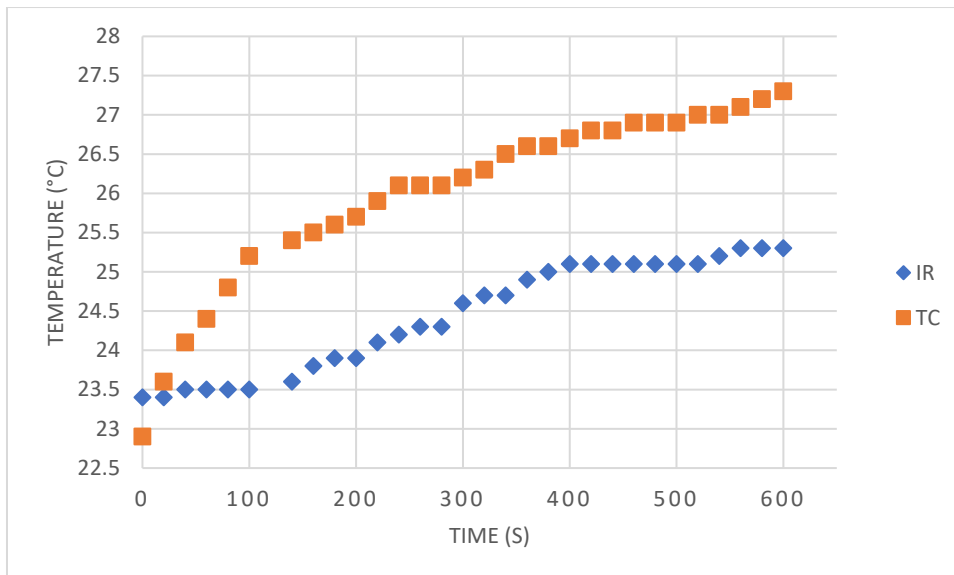


Figure 26 Temperature increase as a function of time in free air with default settings for transmit voltage (30V) and transmit frequency ($f_c=3.6765\text{MHz}$). The PRF is increased to 4 kHz. Thermocouple (TC) and infrared (IR) measurements are shown.

It is shown that there is a discrepancy between the thermocouple and the infrared measurements. The cover material of the C5-2v transducer is reflective, and therefore the infrared thermometer gave measurement results with a high uncertainty. In the next plots, the results obtained with the thermocouple are shown, and will be used to draw conclusions. The temperature increase for this probe in free air using the default transmit settings is limited, only about 5°C in 10 minutes.

Comparing the free air measurements with the test object measurements for the same acquisition in Fig. 27, it was found that the temperature in free air increases faster. However, these measurements would not be the limiting factor considering the higher limits in free air: a temperature increase of 27°C, compared to temperature increase of 10°C in the test object. Therefore, it was decided to do more extensive measurements on the test object.



Figure 27 Temperature increase as a function of time. Default settings for transmit voltage (30V) and center frequency ($f_c=3.6765$ MHz) are used. The PRF was set to 4 kHz. The temperature increase in the two different set-ups are compared.

4.2.2 Test object measurements

The influence of the transmit voltage on the temperature increase was investigated in Fig. 28. For the 40V and 50V measurement, the PRF was increased to 4.5 kHz, which is the maximum achievable PRF for the desired imaging depth of approximately 10 cm.

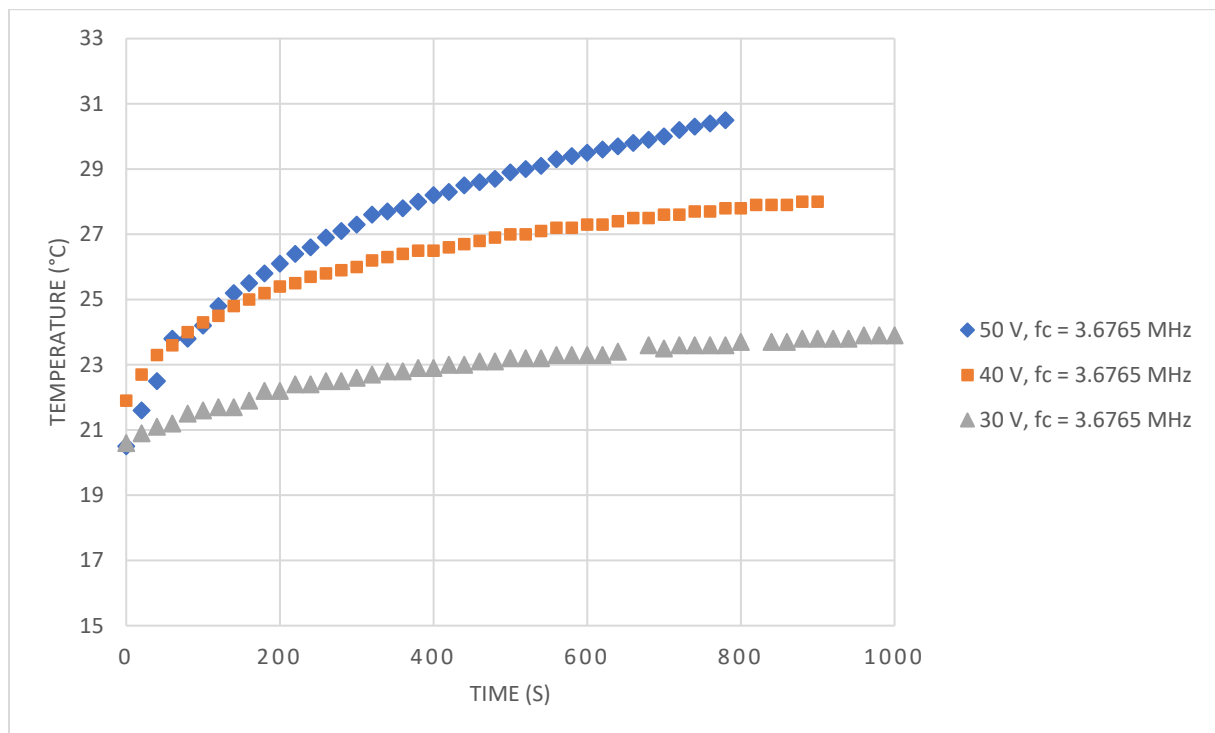


Figure 28 Temperature increase as a function of time. The default center frequency of 3.6765 MHz was used. The transmit voltage was increased from 30V to 50V. The PRF was set to 4 kHz in the 30V measurement, and increased to the maximum achievable PRF of 4.5 kHz in the 40V and 50V measurement.

It was found that using 50V, the temperature limit of a 10°C increase was reached after 13 minutes. The remainder of the measurements were therefore performed with a transmit voltage of 40V, since

using a transmit voltage of 50V would also increase spatial intensity limits, based on the results from the mechanical safety tests. The combination of a transmit voltage of 40 V with a PRF of 4.5 kHz can therefore be regarded as the “worst case scenario”.

The influence of the center frequency was investigated in Fig. 29. Based on the mechanical safety tests, it was found that the optimum center frequency is about 3 MHz. When comparing a center frequency of 3.125 MHz to the default center frequency of 3.6765 MHz, it could be seen that using the optimum center frequency, a higher temperature increase is seen. However, the temperature limit was still not reached after 15 min. The temperature increase after 15 min using a center frequency of 3.125 MHz was equal to 7.5°C.

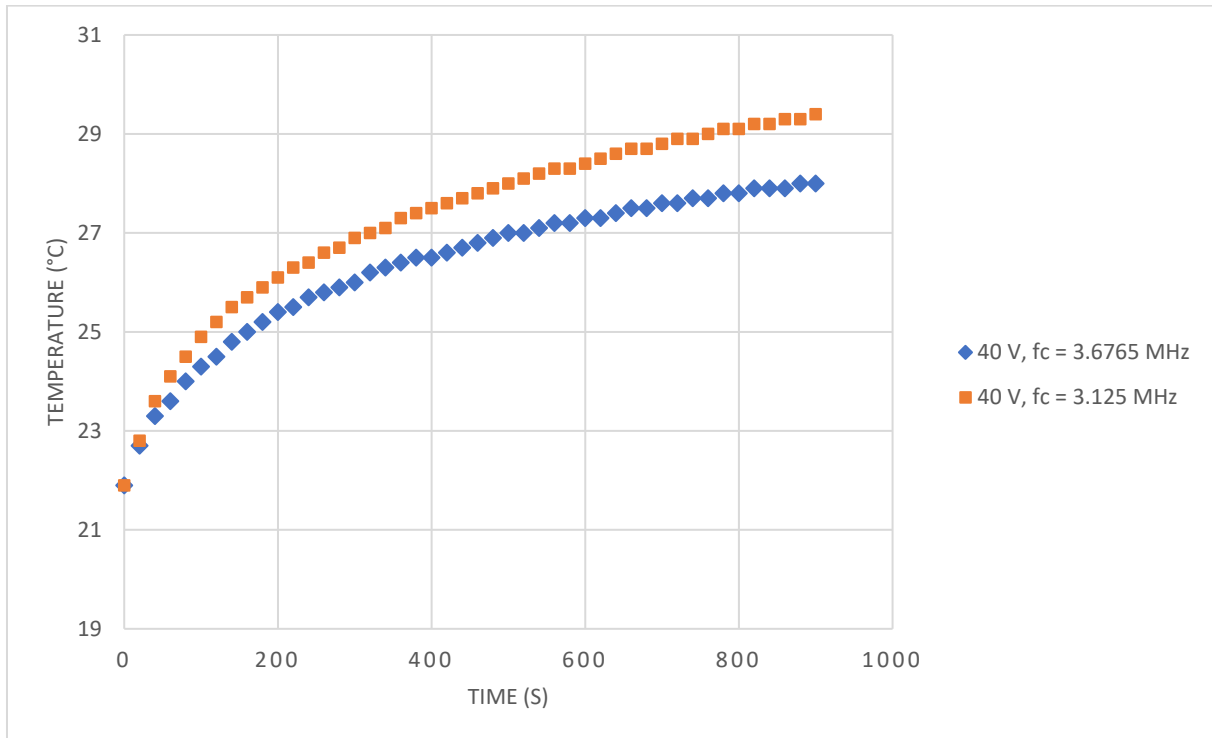


Figure 29 Temperature increase as a function of time. The curve for two different transmit frequencies is shown. The transmit voltage was set to 40 V, and the PRF was equal to 4.5 kHz.

The influence of the transmit angle was investigated in Fig. 30. Here, it is shown that the temperature increase of the transducer surface with a steering angle of 20° is similar to the temperature increase when emitting diverging waves that are not steered.

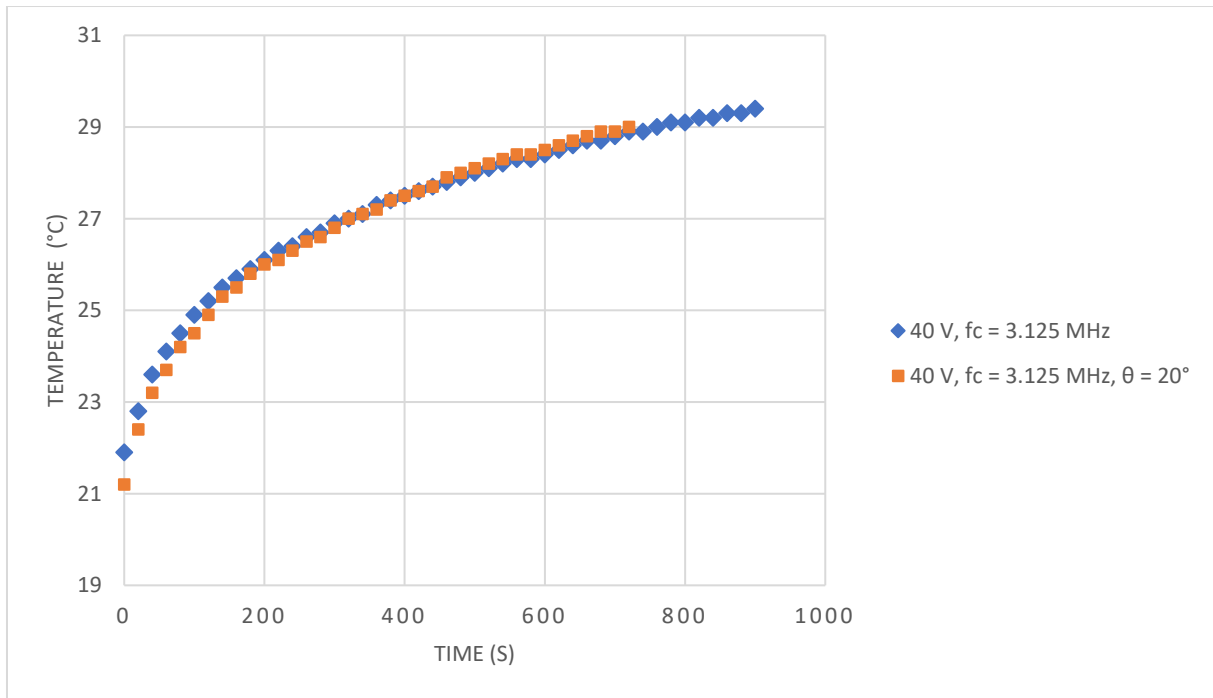


Figure 30 Temperature increase as a function of time. The curve for a 0° steering angle and the curve for a 20° steering angle are shown. The transmit voltage was set to 40 V, the center frequency was 3.125 MHz, and the PRF was equal to 4.5 kHz.

Finally, the two transducers were compared in Fig. 31 for the same acquisition. No differences in temperature increase were found.

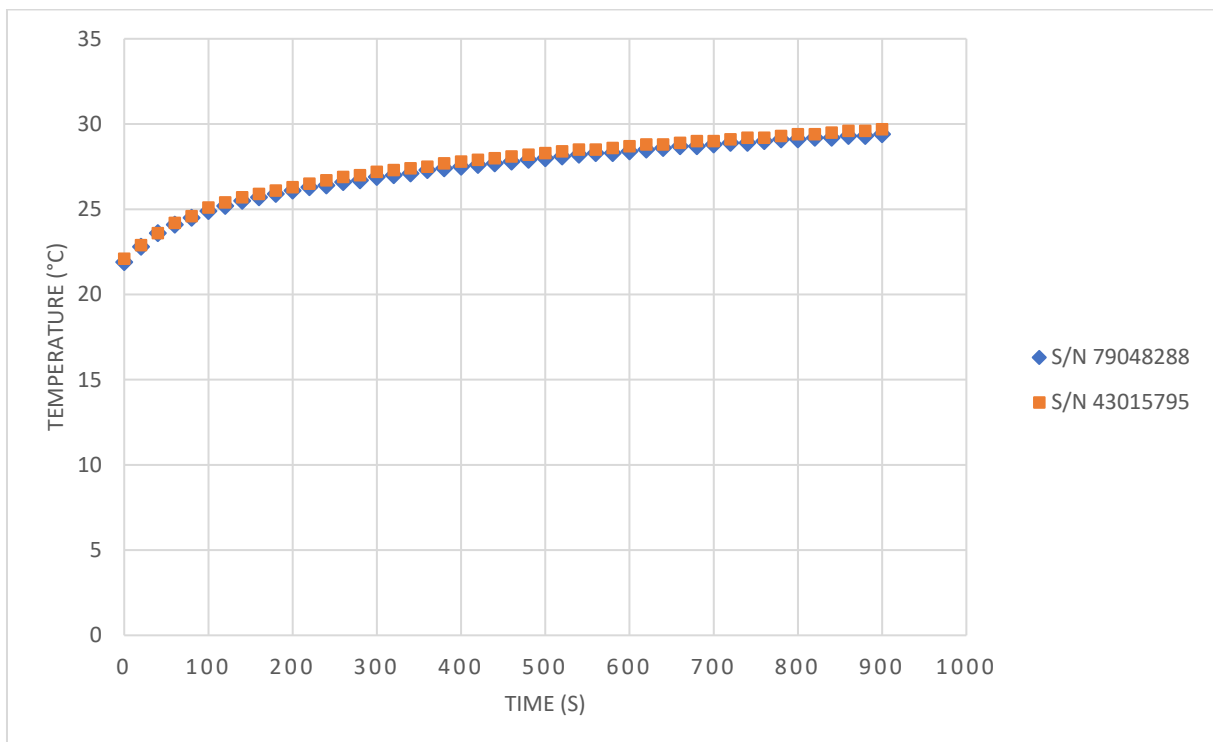


Figure 31 Temperature increase as a function of time. The curve for the same type of curved array transducer with serial numbers 79048288 and 43015795 are shown. The transmit voltage was set to 40 V, the center frequency was 3.125 MHz, and the PRF was equal to 4.5 kHz.

4.2.3 Safety limits

The thermal safety limits as mentioned in Section 2 apply for measurement periods of 30 minutes [2]. The measurements were stopped when this safety limit was reached, or when a steady-state could be observed, e.g. a temperature increase $\leq 0.2^\circ\text{C}/\text{min}$ for 3 consecutive minutes [2]. For all measurements, a steady-state was reached within 15 minutes. At this point, the temperature was still at least 2°C below the set limit of a 10°C temperature increase of the transducer surface. Therefore, it can be concluded that the used acquisition schemes are thermally safe for an unlimited scanning time, as long as the transmit voltage does not exceed 40 V. For 50 V, the set allowed scanning time is equal to 13 minutes. However, this transmit voltage was found to increase spatial intensity limits, and will for that reason not be used.

It should be noted that although the test object set-up was made according to the guidelines in IEC 60601-2-37 (see Section 2), the initial temperature of the test object (21°C) is not comparable to the initial temperature of the human skin (37°C). This means that the surface temperature of the transducer is lower than the temperature of the human skin on which the ultrasound probe is applied. Moreover, human skin will dissipate the heat faster than the used test object. Therefore, it can be expected that the human skin will not warm up significantly during the ultrasound acquisition.

5 Conclusions

The maximum values that are used in this study are:

Neonatal, abdominal, intraoperative, pediatric, small organ, cephalic

	MI	$I_{\text{SPPA.3}}$	$I_{\text{SPTA.3}}$	T_{surface}	$\Delta T_{\text{surface}}$
Limit	1.9	$190 \text{ W}/\text{cm}^2$	$94 \text{ mW}/\text{cm}^2$	43°C (test object) 50°C (free in air)	10°C (test object) 27°C (free in air)

Based on the mechanical and thermal measurements performed, three main conclusions can be drawn regarding the safety of the C5-2v transducer and ultrafast acquisition schemes:

- 1) According to Figure 11, Figure 12 and Table 1 and the spatial intensity limits for MI and $I_{\text{SPPA.3}}$, it was found that the transmit voltage should be limited to 40 V.
- 2) According to Figure 24 and Figure 25, it was found that the maximum allowed PRF is mainly dependent on the transmit voltage used. According to the temporal intensity limit for $I_{\text{SPTA.3}}$, in abdominal applications, desired PRFs of 4 kHz may be applied with transmit voltages up to 20 V. Above 20 V, PRF should be lowered according to Table 6, or the polynomial fit that describes the relationship between voltage and pii_3 (Figure 13). For this study, we expect that the used transmit voltage can be limited to 10 V to obtain good image quality. With this setting, and considering the maximum practically achievable PRF of about 7 kHz, no intensity limits can be exceeded (see Figure 25).
- 3) According to the thermal measurements, there are no limits in the maximum allowed scanning time when transmit voltage is lower than 40 V.

6 Practical implementation of acoustic safety limits

Mechanical and thermal measurements on the transmitted acoustic field were performed in order to in order to design safe clinical studies with human subjects. Settings for transmit voltage, PRF, center frequency and transmit angle in ultrafast imaging with the C5-2v curved array were evaluated. Since the Verasonics research system allows these settings to be manually set, it can be guaranteed that safety limits for abdominal applications will not be exceeded.

To make sure that manually entered settings by the user will not exceed safety limits, a function will be written that checks the combination of these settings. If one of the settings exceeds safety limits, the acquisition script will abort.

Measurements were performed using one curved array transducer. In this study, a dual receive acquisition script will be used in which two curved array transducers receive simultaneously on each transmit event. However, since the transducers alternately transmit, i.e. only one transducer is active at a time, the same safety limits apply. The maximum practically achievable PRF of 7 kHz in abdominal imaging of the aorta also holds for the dual-transducer set-up.

References

- [1] FDA, “Information for manufacturers seeking marketing clearance of diagnostic ultrasound systems and transducers”, *Center for Devices and Radiological Health, United States Food and Drug Administration*, Tech. Rep. 560, 2008 (and draft version 2017)
- [2] IEC 60601:2015, *Medical electrical equipment – Part 2-37: Particular requirements for the basic safety and essential performance of ultrasonic medical diagnostic and monitoring equipment*
- [3] IEC 62359:2010/A1:2017, *Ultrasonics—Field characterization—Test methods for the determination of thermal and mechanical indices related to medical diagnostic ultrasonic fields*
- [4] Precision Acoustics, PPE User Manual. Edition from 31 August 2012. [Available online]: <https://www.acoustics.co.uk/ums-downloads/>
- [5] Onda Hydrophone Handbook. Edition 2. December 2015. [Available online]: <https://ondacorp.com/Handbook/mobile/index.html>
- [6] NEN-EN-IEC 61828:2021, *Ultrasonics - Transducers - Definitions and measurement methods regarding focusing for the transmitted fields* Table H.1: “Speed of sound, c , and characteristic acoustic impedance, pc , as a function of temperature for propagation in water.”

Appendix A Acquisition script Verasonics

Acquisition script used for acoustic output measurements:

```
% Notice:
% This file is provided by Verasonics to end users as a programming
% example for the Verasonics Vantage Research Ultrasound System.
% Verasonics makes no claims as to the functionality or intended
% application of this program and the user assumes all responsibility
% for its use
%
% File name: SetUpC5_2vFlash.m - Example of curved array imaging with flash
%                               transmit
% Description:
% Sequence programming file for C5-2v curved array, using a flash transmit
% and single receive acquisition. All 128 transmit and receive channels
% are active for each acquisition. Processing is asynchronous with
% respect to acquisition.
%
% Testing: Tested with software release 3.0.1
%
% Last update:
% 11-25-2015 Modified for SW 3.0
% 08/19/2015 - Modified for new curved array position with center element at
% origin.
%               Also changed range slider to mm units.
%
% Original file adapted by Vera van Hal for acoustic output
% measurements.

clear

P.startDepth = 5; % P.startDepth is in wavelength
P.endDepth = 250; % P.endDepth is in wavelength

P.numAcqs = 10; %nrframes per superframe
TPC(1).hv = 30.0; % Default Transmit voltage

na = 41; %nr angles in the total sequence
acq_nr = 21; %acquisition number 21 corresponds to 0 degree angle.

if (na > 1)
    dtheta = (40*pi/180)/(na-1);
    startAngle = -20*pi/180;
else
    dtheta = 0;
    startAngle=0;
end % set dtheta to range over +/- 20 degrees.

% Specify system parameters.
Resource.Parameters.numTransmit = 128; % number of transmit channels.
Resource.Parameters.numRcvChannels = 128; % number of receive channels.
Resource.Parameters.speedOfSound = 1480;
Resource.Parameters.connector = 1; % transducer connector to use.
Resource.Parameters.verbose = 2;
Resource.Parameters.initializeOnly = 0;
Resource.Parameters.simulateMode = 0;
% Resource.Parameters.simulateMode = 1 forces simulate mode, even if hardware is
% present.
% Resource.Parameters.simulateMode = 2 stops sequence and processes RcvData
% continuously.

% Specify Trans structure array.
Trans.name = 'C5-2v';
Trans.units = 'wavelengths'; % required in Gen3 to prevent default to mm units
```

```

Trans.frequency = 3.6765;
Trans = computeTrans(Trans); % C5-2 transducer is 'known' transducer so we can use
computeTrans.
Trans.maxHighVoltage = 50; % set maximum high voltage limit for pulser supply.
radius = Trans.radius;
scanangle = Trans.numelements*Trans.spacing/radius;
theta = -(scanangle/2); % angle to left edge from centerline
% Trans.name = 'custom';

% Specify PData structure array.
PData(1).PDelta = [1.0, 0, 0.5]; % x, y and z pdeltas
sizeRows = 10 + ceil((P.endDepth + radius - (radius *
cos(scanangle/2)))/PData(1).PDelta(3));
sizeCols = 10 + ceil(2*(P.endDepth + radius)*sin(scanangle/2)/PData(1).PDelta(1));
PData(1).Size = [sizeRows,sizeCols,1]; % size, origin and pdelta set region of
interest.
PData(1).Origin(1,1) = (P.endDepth+radius)*sin(-scanangle/2) - 5;
PData(1).Origin(1,2) = 0;
PData(1).Origin(1,3) = ceil(radius * cos(scanangle/2)) - radius - 5;
PData(1).Region = struct(...
    'Shape',struct('Name','Sector','Position',[0,0,-
radius],'r1',radius+P.startDepth,'r2',radius+P.endDepth,'angle',scanangle));
PData(1).Region = computeRegions(PData(1));

% Specify Resources.
Resource.RcvBuffer(1).datatype = 'int16';
Resource.RcvBuffer(1).rowsPerFrame = P.numAcqs*4096;
Resource.RcvBuffer(1).colsPerFrame = Resource.Parameters.numRcvChannels;
Resource.RcvBuffer(1).numFrames = 20; % 20 frames for rcvDataLoop buffer.

Resource.InterBuffer(1).datatype = 'complex';
Resource.InterBuffer(1).numFrames = 1; % one intermediate buffer needed.
Resource.InterBuffer(1).pagesPerFrame = 1;

Resource.ImageBuffer(1).datatype = 'double';
Resource.ImageBuffer(1).numFrames = 10;
Resource.DisplayWindow(1).Title = 'C5-2vFlash';
Resource.DisplayWindow(1).numFrames = 200;
Resource.DisplayWindow(1).pdelta = 0.45;
ScrnSize = get(0,'ScreenSize');
DwWidth =
ceil(PData(1).Size(2)*PData(1).PDelta(1)/Resource.DisplayWindow(1).pdelta);
DwHeight =
ceil(PData(1).Size(1)*PData(1).PDelta(3)/Resource.DisplayWindow(1).pdelta);
Resource.DisplayWindow(1).Position = [250,(ScrnSize(4)-(DwHeight+150))/2, ... %
lower left corner position
DwWidth, DwHeight];
Resource.DisplayWindow(1).ReferencePt = [PData(1).Origin(1),0,PData(1).Origin(3)];
% 2D imaging is in the X,Z plane
Resource.DisplayWindow(1).numFrames = 20; %200
Resource.DisplayWindow(1).AxesUnits = 'mm';
Resource.DisplayWindow.Colormap = gray(256);

% Specify Transmit waveform structure.
TW(1).type = 'parametric';
TW(1).Parameters = [Trans.frequency,.67,2,1]; % A, B, C, D

% Specify TX structure array.
TX(1) = repmat(struct('waveform', 1, ...
    'Origin', [0.0,0.0,radius], ...
    'focus', 0.0, ...
    'Steer', [(startAngle+(acq_nr-1)*dtheta),0.0], ...
    'Apod', ones(1,Trans.numelements), ...
    'Delay', zeros(1,Trans.numelements)), 1, 1);

% - Set event specific TX attributes.

TX(1).Apod = tukeywin(Trans.numelements,0.2)';

```

```

TX(1).Delay = computeTXDelays(TX(1));

showTXPD;

% Specify Receive structure arrays.
% -- Compute the maximum receive path length, using the law of cosines.
maxAcqLength = ceil(sqrt((P.endDepth+radius)^2 + radius^2 - ...
    2*(P.endDepth+radius)*radius*cos(scanangle)));
Receive = repmat(struct('Apod', ones(1,Trans.numelements), ...
    'mode', 0, ...
    'startDepth', P.startDepth, ...
    'endDepth', maxAcqLength, ...
    'TGC', 1, ...
    'bufnum', 1, ...
    'framenum', 1, ...
    'acqNum', 1, ...
    'sampleMode', 'NS200BW', ...
    'callMediaFunc', 1), 1, 1,
P.numAcqs*Resource.RcvBuffer(1).numFrames);

% - Set event specific Receive attributes for each frame.
for i = 1:Resource.RcvBuffer(1).numFrames
    %Receive(i).framenum = i;
    for j = 1:P.numAcqs
        Receive(P.numAcqs*(i-1)+j).framenum = i;
        Receive(P.numAcqs*(i-1)+j).acqNum = j;
    end
end

% Specify TGC Waveform structure.
TGC.CntrlPts = [153,308,410,520,605,665,705,760];
TGC.rangeMax = P.endDepth;
TGC.Waveform = computeTGCWaveform(TGC);

% Specify Recon structure arrays.
% - We need one Recon structure which will be used for each frame.
Recon = struct('senscutoff', 0.6, ...
    'pdatanum', 1, ...
    'rcvBufFrame', -1, ...
    'IntBufDest', [1, 1], ...
    'ImgBufDest', [1, -1], ...
    'RINums', 1);

% Define ReconInfo structures.
ReconInfo = struct('mode', 'replaceIntensity', ...
    'txnum', 1, ...
    'rcvnum', 1, ...
    'regionnum', 1);

% Specify Process structure array.
pers = 20;
Process(1).classname = 'Image';
Process(1).method = 'imageDisplay';
Process(1).Parameters = {'imgbufnum', 1, ... % number of buffer to process.
    'framenum', -1, ... % (-1 => lastFrame)
    'pdatanum', 1, ... % number of PData structure to use
    'pgain', 1.0, ... % pgain is image processing gain
    'reject', 2, ... % reject level
    'persistMethod', 'simple', ...
    'persistLevel', pers, ...
    'interpMethod', '4pt', ... %method of interp. (1=4pt)
    'grainRemoval', 'none', ...
    'processMethod', 'none', ...
    'averageMethod', 'none', ...
    'compressMethod', 'power', ...
    'compressFactor', 40, ...
    'mappingMethod', 'full', ...
    'display', 1, ... % display image after processing

```

```

'displayWindow',1};

% Specify SeqControl structure arrays.
SeqControl(1).command = 'jump'; % jump back to start
SeqControl(1).argument = 1;
SeqControl(2).command = 'timeToNextAcq'; % time between frames
SeqControl(2).argument = 1000; % 1 msec
SeqControl(3).command = 'returnToMatlab';
SeqControl(4).command = 'triggerOut'; % this is the trigger that is send with the
acquisition pulse to the hydrophone.
%optional conditions: "syncADC_CLK" synchronizes the trigger out to an edge
%of the specified A/D clock. "syncSYNC_CLC" synchronizes to the SYNC clock,
%with is 1/4 of the A/D clock.
nsc = 5; % nsc is count of SeqControl objects

% Specify Event structure arrays.
n = 1;
nn = 1;
for i = 1:Resource.RcvBuffer(1).numFrames % Acquire frame
    for j=1:P.numAcqs
        Event(n).info = 'Full aperture.';
        Event(n).tx = 1; % use next TX structure.
        Event(n).rcv = nn;
        Event(n).recon = 0; % no reconstruction.
        Event(n).process = 0; % no processing
        Event(n).seqControl = [2,4];
        n = n+1;
        nn = nn+1;
    end
    Event(n-1).seqControl = [2,4,nsc]; % modify last acquisition Event's seqControl
    SeqControl(nsc).command = 'transferToHost'; % transfer frame to host buffer
    nsc = nsc+1;

    Event(n).info = 'recon and process';
    Event(n).tx = 0; % no transmit
    Event(n).rcv = 0; % no rcv
    Event(n).recon = 1; % reconstruction
    Event(n).process = 1; % process
    Event(n).seqControl = 0;
    if (floor(i/5) == i/5) && (i~=Resource.RcvBuffer(1).numFrames) % Exit to Matlab
every 5th frame
        Event(n).seqControl = 3;
    else
        Event(n).seqControl = 0;
    end
    n = n+1;
end

Event(n).info = 'Jump back';
Event(n).tx = 0; % no TX
Event(n).rcv = 0; % no Rcv
Event(n).recon = 0; % no Recon
Event(n).process = 0;
Event(n).seqControl = 1;

% Specify factor for converting sequenceRate to frameRate.
frameRateFactor = P.numAcqs;

% Save all the structures to a .mat file.
save('Vera/C5-2vFlashHydrophone_fastAcq');

%% Automatic VSX Execution:
% Uncomment the following line to automatically run VSX every time you run
% this SetUp script (note that if VSX finds the variable 'filename' in the
% Matlab workspace, it will load and run that file instead of prompting the
% user for the file to be used):

```

```
filename = 'Vera/C5-2vFlashHydrophone_fastAcq'; VSX;  
return;
```